9. 企业综合项目实战

9.1 数据准备

模拟真实的电商网站的数据来统一练习。

1) 建表语句：

```sql
-- 创建用户表
DROP TABLE IF EXISTS ds_hive.ch9_t_member_inf;
create table if not exists ds_hive.ch9_t_member_inf(
 user_id        string COMMENT '用户id'
,idnty_name     string COMMENT '用户姓名'
,idnty_gender   string COMMENT '性别'
,prvnc_name     string COMMENT '省份'
,city_name      string COMMENT '城市'
,rgst_dt        string COMMENT '注册日期'
)
row format delimited fields terminated by '\t'
stored as textfile
;

-- 创建用户分区表
DROP TABLE IF EXISTS ds_hive.ch9_t_member_info_ed;
create table if not exists ds_hive.ch9_t_member_info_ed(
 user_id        string COMMENT '用户id'
,idnty_name     string COMMENT '用户姓名'
,idnty_gender   string COMMENT '性别'
,prvnc_name     string COMMENT '省份'
,city_name      string COMMENT '城市'
,rgst_dt        string COMMENT '注册日期'
)
comment '用户表'
stored as orc
;

-- 创建订单表
DROP TABLE IF EXISTS ds_hive.ch9_t_order;
create table if not exists ds_hive.ch9_t_order(
 order_id         string COMMENT '订单'
,product_id       string COMMENT '产品id'
,user_id          string COMMENT '用户id'
,pay_time         string COMMENT '支付时间(分)'
,amount           string COMMENT '金额'
,status           string COMMENT '状态'
,order_detial     string COMMENT '状态备注'
)
```

```sql
row format delimited fields terminated by '\t'
stored as textfile
;
-- 创建订单分区表
DROP TABLE IF EXISTS ds_hive.ch9_t_order_d;
create table if not exists ds_hive.ch9_t_order_d(
 order_id        string COMMENT '订单'
,product_id      string COMMENT '产品id'
,user_id         string COMMENT '用户id'
,pay_time        string COMMENT '支付时间'
,amount          string COMMENT '金额(分)'
,status          string COMMENT '状态'
,order_detial    string COMMENT '状态备注'
)
comment '订单表'
partitioned by (data_dt string)
stored as orc
;
-- 创建登录表
DROP TABLE IF EXISTS ds_hive.ch9_t_user_login;
create table if not exists ds_hive.ch9_t_user_login(
 login_id        string COMMENT 'id'
,user_id         string COMMENT '用户id'
,login_date      string COMMENT '登录时间'
)
row format delimited fields terminated by '\t'
stored as textfile
;
-- 创建登录分区表
DROP TABLE IF EXISTS ds_hive.ch9_t_user_login_d;
create table if not exists ds_hive.ch9_t_user_login_d(
 login_id        string COMMENT 'id'
,user_id         string COMMENT '用户id'
,login_date      string COMMENT '登录时间'
)
comment '登录表'
partitioned by (data_dt string)
stored as orc
;
```

```sql
-- 创建产品表
DROP TABLE IF EXISTS ds_hive.ch9_t_goods_info;
create table if not exists ds_hive.ch9_t_goods_info(
 goods_id             string COMMENT 'id'
,goods_name           string COMMENT '商品名称'
)
row format delimited fields terminated by '\t'
stored as textfile
;
```

2) 插入数据:

```sql
load data local inpath "/home/hewwen8888/data/t_user.txt"  overwrite into table ds_hive.ch9_t_memb
load data local inpath "/home/hewwen8888/data/t_order.txt"  overwrite into table ds_hive.ch9_t_ord
load data local inpath "/home/hewwen8888/data/t_login.txt"  overwrite into table ds_hive.ch9_t_use
load data local inpath "/home/hewwen8888/data/t_goods.txt"  overwrite into table ds_hive.ch9_t_goo
```

3) 数据清洗

因为订单表和登陆表数据数据量大，必须放到分区表中：

```sql
INSERT OVERWRITE TABLE  ds_hive.ch9_t_member_info_ed
select
 user_id
,idnty_name
,idnty_gender
,prvnc_name
,city_name
,rgst_dt
from ds_hive.ch9_t_member_inf
;

set hive.exec.dynamic.partition.mode=nonstrict; --开启动态分区
INSERT OVERWRITE TABLE  ds_hive.ch9_t_order_d
partition (data_dt)
select
 order_id
,product_id
,user_id
,pay_time
,amount
,status
,order_detial
,regexp_replace(substr(pay_time,1,10),'-','') as data_dt -- 截取年月日数据作为分区依据
from ds_hive.ch9_t_order
;

set hive.exec.dynamic.partition.mode=nonstrict;
INSERT OVERWRITE TABLE  ds_hive.ch9_t_user_login_d
partition (data_dt)
select
 login_id
,user_id
,login_date
,regexp_replace(substr(login_date,1,10),'-','') as data_dt
from   ds_hive.ch9_t_user_login
;
```

9.2 实战练习

0.首先概览所有需要用到的表的结构

ds_hive.ch9_t_member_info_ed

	ch9_t_member_info_ed.user_id	ch9_t_member_info_ed.idnty_name	ch9_t_member_info_ed.idnty_gender	ch9_t_member_info_ed.prvnc_name	ch9_t_member_info_ed.city_name	ch9_t_member_info_ed.rgst_dt
1	0000000000122481184	唐**	1	陕西省	安康市	2024-01-01
2	0000000000151847975	乔**	1	山东省	德州市	2016-07-14
3	0011000001257069210	李**	1	辽宁省	营口市	2019-10-18
4	0000000000299351055	茹**	1	广东省	惠州市	2024-01-01
5	0011000001251849633	黄**	2	山东省	青岛市	2019-09-25
6	0011000001454522521	钟**	1	广西壮族自治区	南宁市	2023-06-20
7	0011000001450644821	胡**	2	湖北省	十堰市	2023-03-05
8	0000000000151523796	张**	1	云南省	文山壮族苗族自治州	2016-07-11
9	0000000000058208894	王**	1	四川省	南充市	2014-07-23
10	0011000001349705817	周**	1	河南省	南阳市	2024-01-01
11	0011000001322779650	赖**	1	江西省	赣州市	2020-05-12
12	0000000000325694495	谢**	2	河北省	承德市	2018-12-01
13	0000000000081065945	华**	1	四川省	南充市	2014-10-30
14	0000000000165476160	张**	1	福建省	莆田市	2016-10-09
15	0011000001412173444	黄**	2	广东省	湛江市	2024-01-01
16	0000000000103499911	周**	1	广东省	惠州市	2024-01-01

ds_hive.ch9_t_order_d

	ch9_t_order_d.order_id	ch9_t_order_d.product_id	ch9_t_order_d.user_id	ch9_t_order_d.pay_time	ch9_t_order_d.amount	ch9_t_order_d.status	ch9_t_order_d.order_detial
1	2312316286152713804	01011000125	0011000001434148883	2024-01-01 00:09:01	300000	S	NULL
2	2401010391154060900	01011000125	0000000000264946752	2024-01-01 10:04:28	354900	S	NULL
3	2401010620153492922	01011000125	0000000000224666581	2024-01-01 01:16:03	699000	S	NULL
4	2401010719153472849	01011000125	0000000000198461020	2024-01-01 03:35:58	1898000	S	NULL
5	2401011866154074766	01011000125	0000000000258503508	2024-01-01 10:38:17	459800	S	NULL
6	2401012254154295733	01011000125	0000000000118834780	2024-01-01 16:31:52	3400000	S	NULL

ds_hive.ch9_t_user_login_d

	ch9_t_user_login_d.order_id	ch9_t_user_login_d.user_id	ch9_t_user_login_d.login_date	ch9_t_user_login_d.data_dt
1	2312316286152713804	0011000001434148883	2024-01-01 00:09:01	20240101
2	2401010391154060900	0000000000264946752	2024-01-01 10:04:28	20240101
3	2401010620153492922	0000000000224666581	2024-01-01 01:16:03	20240101
4	2401010719153472849	0000000000198461020	2024-01-01 03:35:58	20240101
5	2401011866154074766	0000000000258503508	2024-01-01 10:38:17	20240101
6	2401012254154295733	0000000000118834780	2024-01-01 16:31:52	20240101

ds_hive.ch9_t_goods_info

	ch9_t_goods_info.goods_id	ch9_t_goods_info.goods_name
1	01011000125	电视
2	01011000126	洗衣机
3	01011000127	冰箱
4	01011000128	数码相机
5	01011000129	空调
6	01011000130	手机

1. 查询2024年1月份当日注册的并且当天购物的用户数，按每天展示

```
select  substr(t1.pay_time,1,10) as pay_date
         ,count(distinct t1.user_id) as user_cnt
   from ds_hive.ch9_t_order_d t1
   join ds_hive.ch9_t_member_info_ed t2
     on t1.user_id=t2.user_id and substr(t1.pay_time,1,10)=t2.rgst_dt
where t1.data_dt>='20240101'  and t1.data_dt<='20240131'
 group by substr(t1.pay_time,1,10)
 ;
```

执行结果:


查询历史记录 保存的查询 查询生成器 结果 (3)

	pay_date	user_cnt
1	2024-01-01	9
2	2024-01-02	2
3	2024-01-03	2

2. 查询2024年一月份，累积销量排名前三的商品和对应的销售量

```
select
t4.goods_name
,t4.user_cnt
from
(select
 t3.goods_name
 ,t3.user_cnt
 ,row_number() over(order by user_cnt desc ) as num
 from
(
 select  t2.goods_name
         ,count(distinct t1.order_id) as user_cnt
   from ds_hive.ch9_t_order_d t1
   join  ds_hive.ch9_t_goods_info t2
     on t1.product_id=t2.goods_id
where t1.data_dt>='20240101'  and t1.data_dt<='20240131'
 group by t2.goods_name
 )
 t3
 )
 t4
 where t4.num<=3
 ;
```

t3:

	t2.goods_name	user_cnt
1	冰箱	103
2	手机	21
3	数码相机	76
4	洗衣机	60
5	电视	55
6	空调	64

t4:

	t3.goods_name	t3.user_cnt	num
1	冰箱	103	1
2	数码相机	76	2
3	空调	64	3
4	洗衣机	60	4
5	电视	55	5
6	手机	21	6

最终结果:

查询历史记录 保存的查询 查询生成器 结果 (3)

	t2.goods_name	user_cnt
1	冰箱	103
2	数码相机	76
3	空调	64

3. 查询至少连续三天下单的用户

```
------第一步，保障每天只有一条数据
with  ch9_timu3_order_01 as (
 select  substr(t1.pay_time,1,10) as pay_date
         ,t1.user_id
         ,count(*) as cnt
   from ds_hive.ch9_t_order_d t1
where t1.data_dt is not null
 group by substr(t1.pay_time,1,10) ,t1.user_id
),
----------第二步，构建等差相减
ch9_timu3_order_02 as (
  select user_id
         ,num
         ,count(*) as cnt
  from
  (select user_id
        ,pay_date
        ,date_sub(pay_date,row_number() over(partition by user_id order by pay_date)) num
   from ch9_timu3_order_01
   ) t1
   group  by
   user_id
         ,num
having count(*)>=3
 )
 select user_id from ch9_timu3_order_02
 ;
```

ch9_timu3_order_01:

	pay_date	t1.user_id	cnt
1	2024-01-01	0000000000018581106	1
2	2024-01-01	0000000000018620986	1
3	2024-01-01	0000000000021128303	1
4	2024-01-01	0000000000037245020	1
5	2024-01-01	0000000000058208894	1
6	2024-01-01	0000000000066826842	1
7	2024-01-01	0000000000080937692	3
8	2024-01-01	0000000000081065945	1
9	2024-01-01	0000000000088880096	3

ch9_timu3_order_02:

	user_id	num	cnt
1	0000000000018581106	2023-12-31	3
2	0000000000018620986	2023-12-31	3
3	0000000000021128303	2023-12-31	3
4	0000000000058208894	2023-12-31	3
5	0000000000080937692	2023-12-31	5
6	0000000000088880096	2023-12-31	5
7	0000000000095704339	2023-12-31	5
8	0000000000101777738	2023-12-31	5

最终结果:

	user_id
1	0000000000018581106
2	0000000000018620986
3	0000000000021128303
4	0000000000058208894
5	0000000000080937692
6	0000000000088880096
7	0000000000095704339
8	0000000000101777738
9	0000000000103499911
10	0000000000109627255
11	0000000000111269038
12	0000000000122481184
13	0000000000125441929
14	0000000000128064012
15	0000000000141404385
16	0000000000142325925

4. 查询用户的累计消费金额及根据销售金额算出用户等级

用户vip等级根据累积消费金额计算，计算规则如下：
设累积消费总额为X，
若0=<X<2000,则vip等级为青铜会员
若2000<=X<3000,则vip等级为白银会员
若3000<=X<5000,则vip等级为黄金会员
若5000<=X<8000,则vip为钻石会员

若8000<=X<10000,则vip等级为星耀会员
若X>=10000,则vip等级为王者会员

```
with ch9_timu4_01 as (
    select user_id
           ,pay_date
           ,amount
           ,sum(amount) over(partition by user_id order by pay_date) as sum_amount
    from (
        -- 每个用户每天的金额
        select substr(t1.pay_time,1,10) as pay_date
               ,t1.user_id
               ,sum(amount) as amount
        from ds_hive.ch9_t_order_d t1
        where t1.data_dt is not null
        group by substr(t1.pay_time,1,10), t1.user_id
    ) t2
),
user_total as (
    -- 计算每个用户的总消费额（即最大累计金额）
    select user_id, max(sum_amount) as total_amount
    from ch9_timu4_01
    group by user_id
)
select user_id
       ,total_amount
       ,case when 0 <= total_amount and total_amount < 2000 then '青铜会员'
            when 2000 <= total_amount and total_amount < 3000 then '白银会员'
            when 3000 <= total_amount and total_amount < 5000 then '黄金会员'
            when 5000 <= total_amount and total_amount < 8000 then '钻石会员'
            when 8000 <= total_amount and total_amount < 10000 then '星耀会员'
            else '王者会员'
        end as level
from user_total;
```

ch9_timu4_01:

	user_id	pay_date	amount	sum_amount
1	0000000000018581106	2024-01-01	479900	479900
2	0000000000018581106	2024-01-02	479900	959800
3	0000000000018581106	2024-01-03	479900	1439700
4	0000000000018581106	2024-01-05	479900	1919600
5	0000000000018620986	2024-01-01	153700	153700
6	0000000000018620986	2024-01-02	153700	307400
7	0000000000018620986	2024-01-03	153700	461100
8	0000000000018620986	2024-01-05	153700	614800

最终结果：

	user_id	total_amount	level
1	0000000000018581106	1919600	王者会员
2	0000000000018620986	614800	王者会员
3	0000000000021128303	1380536	王者会员
4	0000000000037245020	559600	王者会员
5	0000000000058208894	18000000	王者会员
6	0000000000066826842	20700	钻石会员
7	0000000000080937692	2219200	王者会员
8	0000000000081065945	1559400	王者会员

5. 查询2024年1月份每个用户，每天购买的商品，按单个用户展现

```
select   t4.pay_date
          ,t4.idnty_name
          ,concat_ws('|',collect_set( t4.goods_name)) goods_name
 from
 (select  substr(t1.pay_time,1,10) as pay_date
         ,idnty_name
         ,goods_name
         ,count(*) as cnt
   from ds_hive.ch9_t_order_d t1
   join ds_hive.ch9_t_member_info_ed t2
     on t1.user_id=t2.user_id
   join  ds_hive.ch9_t_goods_info t3
     on t1.product_id=t3.goods_id
where t1.data_dt>='20240101'  and t1.data_dt<='20240131'
 group by substr(t1.pay_time,1,10)
         ,idnty_name
         ,goods_name
 ) t4
 group by
 t4.pay_date
          ,t4.idnty_name
 ;
```

t4:

	pay_date	idnty_name	goods_name	cnt
1	2024-01-01	乔**	冰箱	2
2	2024-01-01	任**	手机	1
3	2024-01-01	何**	手机	2
4	2024-01-01	储**	冰箱	1
5	2024-01-01	刘**	数码相机	1
6	2024-01-01	刘**	电视	1
7	2024-01-01	华**	电视	1
8	2024-01-01	司**	电视	1

最终结果：

	t4.pay_date	t4.idnty_name	goods_name
1	2024-01-01	乔**	冰箱
2	2024-01-01	任**	手机
3	2024-01-01	何**	手机
4	2024-01-01	储**	冰箱
5	2024-01-01	刘**	数码相机|电视
6	2024-01-01	华**	电视
7	2024-01-01	司**	电视
8	2024-01-01	吴**	洗衣机|空调

6. 统计2024年1月份每个商品的销金额最高的日期，如果销售金额相同，取最小日期

```
select
t3.product_id
,t3.pay_date
,t3.amount
from
(select
   product_id
   ,pay_date
   ,amount
   ,row_number() over(partition by product_id order by amount desc,pay_date asc) as num
   from
(
select  substr(t1.pay_time,1,10) as pay_date
         ,product_id
         ,sum(amount) as amount
   from ds_hive.ch9_t_order_d t1
where t1.data_dt>='20240101'  and t1.data_dt<='20240131'
group  by
substr(t1.pay_time,1,10)
         ,product_id
) t2
) t3
where t3.num=1
;
```

t2:

	pay_date	product_id	amount
1	2024-01-01	01011000125	13085026
2	2024-01-01	01011000126	10775161
3	2024-01-01	01011000127	17164030
4	2024-01-01	01011000128	16496655
5	2024-01-01	01011000129	12129725
6	2024-01-01	01011000130	6342000
7	2024-01-02	01011000125	3643526
8	2024-01-02	01011000126	10775161

t3:

	product_id	pay_date	amount	num
1	01011000125	2024-01-04	13244826	1
2	01011000125	2024-01-01	13085026	2
3	01011000125	2024-01-05	9373326	3
4	01011000125	2024-01-02	3643526	4
5	01011000126	2024-01-01	10775161	1
6	01011000126	2024-01-02	10775161	2
7	01011000126	2024-01-04	10775161	3
8	01011000126	2024-01-05	10775161	4

最终结果：

	t3.product_id	t3.pay_date	t3.amount
1	01011000125	2024-01-04	13244826
2	01011000126	2024-01-01	10775161
3	01011000127	2024-01-01	17164030
4	01011000128	2024-01-01	16496655
5	01011000129	2024-01-01	12129725
6	01011000130	2024-01-01	6342000

7. 查询2024年1月份所有用户的连续登录三天及以上的日期区间

```
with timu7_table_tmp1 as(
select t1.user_id
,substr(login_date,1,10) as login_date
from ds_hive.ch9_t_user_login_d t1
where t1.data_dt is not null
group by substr(login_date,1,10),t1.user_id
)
select
t3.user_id
,min(t3.login_date) as min_login_date
,max(t3.login_date) as max_login_date
from
(
select user_id
,login_date
,date_add(login_date,row_number() over(partition by user_id order by login_date desc)) as dt_flag
from timu7_table_tmp1
) t3
group by t3.user_id,t3.dt_flag
having count(*)>=3
;
```

timu7_table_tmp1：

	t1.user_id	login_date
1	0000000000018581106	2024-01-01
2	0000000000018620986	2024-01-01
3	0000000000021128303	2024-01-01
4	0000000000037245020	2024-01-01
5	0000000000058208894	2024-01-01
6	0000000000066826842	2024-01-01
7	0000000000080937692	2024-01-01
8	0000000000081065945	2024-01-01

t3:

	user_id	login_date	dt_flag
1	0000000000018581106	2024-01-05	2024-01-06
2	0000000000018581106	2024-01-03	2024-01-05
3	0000000000018581106	2024-01-02	2024-01-05
4	0000000000018581106	2024-01-01	2024-01-05
5	0000000000018620986	2024-01-05	2024-01-06
6	0000000000018620986	2024-01-03	2024-01-05
7	0000000000018620986	2024-01-02	2024-01-05
8	0000000000018620986	2024-01-01	2024-01-05

最终结果：

询历史记录　保存的查询　查询生成器　结果 (50)

	user_id	begin_dt	end_dt
1	0000000000018581106	2024-01-01	2024-01-03
2	0000000000018620986	2024-01-01	2024-01-03
3	0000000000021128303	2024-01-01	2024-01-03
4	0000000000058208894	2024-01-01	2024-01-03
5	0000000000080937692	2024-01-01	2024-01-05
6	0000000000088880096	2024-01-01	2024-01-05
7	0000000000095704339	2024-01-01	2024-01-05
8	0000000000101777738	2024-01-01	2024-01-05
9	0000000000103499911	2024-01-01	2024-01-03
10	0000000000109627255	2024-01-01	2024-01-05

8. 查询2024年1月每个用户的最近一笔订单的日期和金额

```
select
 user_id
 ,Order_id
 ,pay_date
 ,amount
 from
(
 select  substr(t1.pay_time,1,10) as pay_date
         ,t1.user_id
         ,t1.Order_id
         ,t1.amount
         ,row_number() over(partition by user_id order by pay_time  desc) num
   from ds_hive.ch9_t_order_d t1
where t1.data_dt>='20240101'  and t1.data_dt<='20240131'
)
t1
where num=1
;
```

t1:

	pay_date	t1.user_id	t1.order_id	t1.amount	num
1	2024-01-05	0000000000018581106	2401055808154560230	479900	1
2	2024-01-03	0000000000018581106	2401035808154560230	479900	2
3	2024-01-02	0000000000018581106	2401025808154560230	479900	3
4	2024-01-01	0000000000018581106	2401015808154560230	479900	4
5	2024-01-05	0000000000018620986	2401059638153508413	153700	1
6	2024-01-03	0000000000018620986	2401039638153508413	153700	2
7	2024-01-02	0000000000018620986	2401029638153508413	153700	3
8	2024-01-01	0000000000018620986	2401019638153508413	153700	4

最终结果:

	user_id	order_id	pay_date	amount
1	0000000000018581106	2401055808154560230	2024-01-05	479900
2	0000000000018620986	2401059638153508413	2024-01-05	153700
3	0000000000021128303	2401056078154521984	2024-01-05	345134
4	0000000000037245020	2401057903154451850	2024-01-05	139900
5	0000000000058208894	2401058156154115564	2024-01-05	4500000
6	0000000000066826842	2401023338154501232	2024-01-05	6900
7	0000000000080937692	2401057382154259857	2024-01-05	110000
8	0000000000081065945	2401022348153852199	2024-01-05	519800
9	0000000000088880096	2401058925153425696	2024-01-05	150000
10	0000000000095704339	2401052762153437307	2024-01-05	246777
11	0000000000101394445	2401023113154418479	2024-01-05	479900
12	0000000000101777728	2401054003154266782	2024-01-05	689700

9. 查询查询2024年1月每个产品每天的历史累计金额总和以及历史累计日平均值

```
with timu9_table_tmp1 as(
select t1.product_id
,substr(pay_time,1,10) as order_dt
,cast(sum(amount/100) as decimal(10,2)) as goods_amt
from ds_hive.ch9_t_order_d t1
where t1.data_dt<='20240131'
group by substr(pay_time,1,10),t1.product_id
)
select product_id
,order_dt
,sum(goods_amt) over(partition by product_id order by order_dt asc) as sum_all
,avg(goods_amt) over(partition by product_id order by order_dt asc) as avg_all
from timu9_table_tmp1
;
```

timu9_table_tmp1：

	t1.product_id	order_dt	goods_amt
1	01011000125	2024-01-01	130850.26
2	01011000126	2024-01-01	107751.61
3	01011000127	2024-01-01	171640.30
4	01011000128	2024-01-01	164966.55
5	01011000129	2024-01-01	121297.25
6	01011000130	2024-01-01	63420.00
7	01011000125	2024-01-02	36435.26
8	01011000126	2024-01-02	107751.61

最终结果：

	product_id	order_dt	sum_all	avg_all
1	01011000125	2024-01-01	130850.26	130850.260000
2	01011000125	2024-01-02	167285.52	83642.760000
3	01011000125	2024-01-04	299733.78	99911.260000
4	01011000125	2024-01-05	393467.04	98366.760000
5	01011000126	2024-01-01	107751.61	107751.610000
6	01011000126	2024-01-02	215503.22	107751.610000
7	01011000126	2024-01-04	323254.83	107751.610000
8	01011000126	2024-01-05	431006.44	107751.610000

10. 查询每天的首购和复购用户数，首购是第一次购买商品，复购是第二次购物或者二次以上购物

```
select
 t3.all_dt   as pay_dt
 ,count(case when t3.all_dt=t4.First_dt  then   t3.user_id end )  as First_dt_cnt
 ,count(case when t3.all_dt<>t4.First_dt  then  t3.user_id end )  as last_dt_cnt
 from
(
select  user_id
        ,substr(t1.pay_time,1,10)  as all_dt
   from ds_hive.ch9_t_order_d t1
where t1.data_dt>='20240101'  and t1.data_dt<='20240131'
group   by user_id
           ,substr(t1.pay_time,1,10)
) t3
left join
(select  user_id
        ,min(substr(t1.pay_time,1,10) )  as First_dt
   from ds_hive.ch9_t_order_d t1
where t1.data_dt>='20240101'  and t1.data_dt<='20240131'
group   by user_id
) t4
on t3.user_id=t4.user_id
group by t3.all_dt
;
```

t3:

	user_id	all_dt
1	0000000000018581106	2024-01-01
2	0000000000018581106	2024-01-02
3	0000000000018581106	2024-01-03
4	0000000000018581106	2024-01-05
5	0000000000018620986	2024-01-01
6	0000000000018620986	2024-01-02
7	0000000000018620986	2024-01-03
8	0000000000018620986	2024-01-05

t4:

	user_id	first_dt
1	0000000000018581106	2024-01-01
2	0000000000018620986	2024-01-01
3	0000000000021128303	2024-01-01
4	0000000000037245020	2024-01-01
5	0000000000058208894	2024-01-01
6	0000000000066826842	2024-01-01
7	0000000000080937692	2024-01-01
8	0000000000081065945	2024-01-01

最终结果：

	pay_dt	first_dt_cnt	last_dt_cnt
1	2024-01-01	74	0
2	2024-01-02	0	63
3	2024-01-03	0	52
4	2024-01-04	0	41
5	2024-01-05	0	69